

# Homework for Introduction to Mathematics

1. Solve the following questions with the specified method.

- (a) Negate  $(P \Rightarrow Q) \Rightarrow P$ , you don't need to simplify it.
- (b) Using  $((P \wedge Q) \vee P) \equiv P$  to justify  $\sim [(P \Rightarrow Q) \Rightarrow P]$  and  $\sim P$  are logically equivalent.
- (c) State the definition of logically equivalent, and using it to justify

$$((P \Rightarrow Q) \Rightarrow P) \Leftrightarrow P$$

is tautology.

2. For the following statement forms, write down their logically equivalent statement forms. **Only using  $\sim$  and  $\vee$ .**

- (a)  $P \Leftrightarrow (\sim Q)$ .
- (b)  $(P \vee Q) \Rightarrow (P \wedge Q)$ .
- (c)  $(P \Rightarrow Q) \vee (Q \Rightarrow P)$ .

3. Let  $P$  be the statement: If  $x^2 - 3x + 2 < 0$ , then  $1.3 < x < 1.7$ . Write  $\sim P$ .